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ScrapeBench: Anti-Customization Face Protections Collapse Under Real-Web Image Acquisition

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ABSTRACT

Protections that promise to make personal photographs unlearnable by unauthorized diffusion-based customization are forged and validated in a laboratory: clean full-resolution files, lossless pipelines, and a single tunable fine-tuning attacker. Images scraped from the web live in a different world, and the distance between the two worlds is the subject of this paper. We name it the *Web Acquisition Gap*: the stack of lossy transformations that lies between the file a defender protects and the copy an adversary actually trains on—platform transcoding, browser and thumbnail rendering, agent-side capture, restoration, and diffusion-based regeneration. We present *ScrapeBench*, the first evaluation harness that treats this acquisition funnel as a first-class, tiered attack surface: seven tiers with explicit per-image attacker budgets, whose parameters are anchored not in speculation but in measured platform behavior (GitHub avatar re-encoding at JPEG quality 75, Gravatar at 90, WordPress Photon at 80, image proxies exposing the whole 60–90 band). ScrapeBench reports a *retention ratio* $\mathrm{RR}(T)$—the fraction of a defense's clean-domain protection gap that survives at tier T—with paired significance tests, an integrated funnel protection score, and an explicit separation between *defense bypass* and *image destruction*. Across 40 public identities, four T_0-optimized perturbation defenses and two controls, three attack families (DreamBooth-LoRA, zero-shot IP-Adapter-FaceID, and InstantID), we find the gap is one-sided and structural: the platform's own tiers erase most of the T0 protection gap before any adaptive attacker acts; the order and composition of transformations do not rescue the defenses; a JPEG-aware retrained variant recovers part of the transcoding tier yet remains exposed to rendering and regeneration; and modest data-level adaptation closes what remains, while the cheapest tiers convert some defenses into image destruction rather than protection. Forging costs GPU-minutes per identity; defeating costs milliseconds per image, chosen by parties with no stake in the attack. We complement the benchmark with `face-consent.txt`, a robots.txt-style machine-readable consent protocol with a reference validator, and argue that durable protection is an acquisition-layer systems problem, not a stronger-perturbation problem. We release the harness, platform measurements, consent prototype, and all curves.

1 INTRODUCTION

The web has become the default training corpus for identity-hungry generative models, and faces are its most sensitive commodity. A handful of public portraits now suffices to synthesize a person in any context: DreamBooth-style fine-tuning binds a rare token to an identity with four to eight images [33], while zero-shot adapters such as InstantID and IP-Adapter-FaceID inject a face embedding from a single reference image directly into the denoising process [44, 52]. The consent asymmetry is severe—the subject never agreed—and it has triggered an arms race in *protective perturbation*: cloaks that, added to a photo before upload, break the customization pipeline. Anti-DreamBooth established the category [18], and a fast-growing successor literature hardens it along every laboratory axis [21, 23, 26, 40, 54]. The narrative is reassuring: with the right perturbation, your face is yours again.

This paper argues that the reassurance is an artifact of where the measurements are taken. Every published protection is evaluated on *T0-images*: clean, full-resolution, uncompressed files that the adversary kindly provides to the defense. That world does not exist on the web. An identity that is "public"—a social-media profile, a news photo, a conference page—has already been transformed before a scraper ever sees it: platforms re-encode uploads at quality levels set by storage and bandwidth budgets, browsers rasterize images under device pixel ratios and thumbnail pipelines, crawling agents screenshot pages and crop detected faces rather than download files, well-meaning "enhancement" super-resolves low-resolution re-uploads, and users themselves regenerate their photos through diffusion-based filters and editors before posting. Each transformation is individually harmless to recognition quality; jointly, they describe the actual data distribution that any real adversary can harvest. We call this pipeline the *web acquisition funnel*, and the resulting evaluation blind spot the *Web Acquisition Gap*: the systematic difference between the protection a defense delivers on the files it was handed and the protection that survives on the files an adversary can actually obtain. The funnel, we claim, is not a nuisance variable for anti-customization protections but their primary adversary—an attack surface that requires no adversarial optimization whatsoever, costs milliseconds to seconds per image on commodity platforms, and operates *before* the defense is ever consulted.

Two observations sharpen the claim. First, the funnel is not hypothetical: published measurements already show that a single JPEG re-encode at quality 85–95 neutralizes a broad class of protection perturbations [35], that common image transformations systematically suppress unlearnable-example attacks [30, 36], and that robustness evaluations of protective perturbations under diffusion-aware adversaries remain unfavorable [29]. Nor is the funnel's parameter range speculative: we probe production image pipelines directly and find GitHub's avatar service re-encoding at JPEG quality 75, Gravatar at 90, and WordPress's Photon CDN at 80—operating points that fall squarely inside, not at the edge of, the transcode tiers we evaluate. What is missing is a systematic, tiered, budget-aware account of the entire funnel—from platform transcode to agent capture to restoration to regeneration—and of *where along the funnel* each family of defenses actually collapses, measured under both tunable and zero-shot attack pipelines. Second, the evaluation practices of the field are drifting in the opposite direction: newest defenses arrive with evaluations on ever-cleaner galleries under ever-stronger laboratory attackers [2, 5], and the first cross-method

evaluation studies, while valuable, still hold the acquisition side fixed [43]. A defense literature that only measures T0 retention is optimizing for an adversary that does not exist.

We build the missing instrument. *ScrapeBench* reconstructs the funnel in software: seven tiers (T0 original; T1 platform transcode; T2 browser and thumbnail rendering; T3 agent-style capture with face cropping; T4 super-resolution restoration; T5 diffusion regeneration; T6 adaptive mixing and purification), each with an explicit attacker budget. On top of it we define the *retention ratio* $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$, where $\Delta(T)$ is the protection gap (clean minus protected identity similarity under a fixed attack) at tier T: RR measures the fraction of a defense's headline T0 protection that survives acquisition reality. The metric is deliberately paired with a cost ledger, because the funnel's defining property is an *inversion of asymmetry*—the defender pays GPU-minutes per identity to forge a perturbation, the attacker pays CPU-milliseconds per image to defeat it.

Running ScrapeBench over 40 public identities, four defenses spanning the protection design space (clean control, random noise, FSMG-style untargeted surrogate attacks [18], and ASPL-lite alternation) plus a JPEG-aware retrained variant that stress-tests the "just make it JPEG-robust" fix, and three attack families (DreamBooth-LoRA fine-tuning and the zero-shot adapters IP-Adapter-FaceID and InstantID) yields four findings, each readable from the collapse curves in Section 5: **(F1)** protection sheds most of its gap within the *platform-controlled* tiers (T1–T2), before any adaptive attacker enters; **(F2)** zero-shot identity adapters, which skip fine-tuning entirely, erase most of what survives the platform tiers, because the perturbations were never optimized against embedding injection; **(F3)** the cost asymmetry runs against the defender by orders of magnitude, converting "protection" into a tax the attacker routes around; **(F4)** the failure is structural, not parametric—reordering or composing the funnel's transformations does not restore protection, and the JPEG-aware variant that recovers part of the transcode tier stays exposed to rendering and regeneration. Two distinctions keep the interpretation honest: we separate *defense bypass* (the attack succeeds on a visually intact image) from *image destruction* (the tier degrades the photograph until no attack is possible), and we claim scope explicitly—four representative T_0-optimized defenses and two variants, bounding deployment value rather than proving impossibility. The constructive turn is a funnel-aware reporting standard (publish RR@T, not T0-only numbers) and a consent layer that survives the funnel by construction: `face-consent.txt`, a robots.txt-style consent manifest with a reference validator.

Our contributions are:

- **The Web Acquisition Gap and a funnel threat model, anchored in platform reality.** We formalize the gap between laboratory protection evaluation and web acquisition as a seven-tier funnel with per-tier attacker budgets, and instrument public image services (GitHub avatars, Gravatar, WordPress Photon, an open proxy) whose re-encoding and resizing behavior covers exactly the tiers that collapse the defenses—making the threat model empirical, not hypothetical.
- **The retention-ratio metric and collapse-curve protocol.** $\mathrm{RR}(T)$ reduces robustness under acquisition to one interpretable curve per defense; a cost ledger exposes the defender–attacker asymmetry that raw numbers hide; an infeasibility view separates defense bypass from image destruction.
- **ScrapeBench, an open harness.** A reproducible pipeline—transcoding, browser/agent rendering, restoration, regeneration, adaptive pooling—integrated with DreamBooth-LoRA and two zero-shot attackers, ArcFace scoring, paired statistics, and automatic result publication.
- **Empirical collapse findings with stress tests.** Across 40 identities, four defenses, two controls, a JPEG-aware variant, and three attack families we quantify F1–F5, including order-and-composition ablations—establishing where, not just whether, protections fail.
- **A consent layer that survives the funnel.** `face-consent.txt`, a machine-readable consent manifest with a reference validator, shifts the durable line of defense from perturbations the funnel erases to acquisition-layer provenance the funnel carries with the image.

Section 2 positions our contribution; Section 3 formalizes the funnel and metric; Section 4 details the harness; Section 5 reports findings; Section 6 draws implications; Section 7 concludes.

2 RELATED WORK

Unauthorized customization and its defenders. Personalization of text-to-image diffusion models has collapsed the cost of fabricating a person's likeness: DreamBooth binds a subject to a rare token through a handful of images [33], textual inversion and multi-concept customization extend the mechanism to arbitrary subjects [7, 17], latent diffusion provides the efficient backbone that made such fine-tuning ubiquitous [32], and zero-shot identity adapters skip fine-tuning entirely by injecting face-recognition embeddings through decoupled cross-attention [52] or identity-preserving pipelines [44], with the design space now surveyed at scale [1]. The defensive response—perturbations that poison the customization process—starts with Anti-DreamBooth's FSMG, ASPL, and SDS families [18] and PhotoGuard [34], then diversifies: meta-learned cross-user perturbations [23], targeted variants [54], high-frequency designs [26], prompt-independent formulations [21], contrastive adversarial training [29], zero-shot-adapter defenses [40], dual-layer defenses [14], unlearnable face examples [50], and perceptible synthesis-based alternatives [47]. The 2025–2026 wave broadens along three fronts. VOID hardens latent-diffusion mimicry against prompt-injection and gradient-collision adversaries [31]; the Anti-Aesthetics line replaces imperceptible perturbation with engineered perceptual degradation, first against text-to-image customization [45] and then as a hierarchical anti-aesthetic curriculum [46]; and purification-robust formulations explicitly anticipate attackers who launder protected galleries through diffusion regeneration [51]. Parallel work pushes into video protection [9, 48], face-swap frameworks [16], face-shield embeddings against deepfake manipulation [13], and JPEG-robust formulations [5]. Each of these defenses is designed against a *named* adversary—an injection mechanism, human perception, or a single purification operator—yet none evaluates whether the protection survives the unmanaged acquisition path between a real platform and the attacker's training

script; the funnel tiers of Section 3 are precisely that unmodeled surface. Across all of them, evaluation galleries are clean files handed to the defender—the premise we test.

The fragility literature. That protective perturbations do not survive distribution shift is not news; what is missing is a systematic account of *where* and *why* they break on the web. Sandoval-Segura et al. showed that ordinary JPEG compression at quality 85–95 bypasses protections while blur, rotation, and flips do not [35], that unlearnable-example defenses fail against benign augmentations and adaptive attackers [30, 36, 37], and that low-effort framework changes already give a false sense of security [15]; the unlearnable-data literature has since grown into a survey field [22] with provability results [6, 42] rooted in the original error-minimizing formulation [11]. Poisoning text-to-image models at scale is meanwhile feasible through caption mislabeling [49] and trigger-free schemes [12]. Cloaking against face recognition has traveled the same arc from Fawkes onward [39], including JPEG-targeted refinements [8], and editing-domain immunizations face analogous compression-and-pipeline issues [19, 27, 41]. In the adjacent artist-protection domain, Glaze spawned an evaluation arms race in which adversarial perturbations proved unreliable against simple transformations [10, 38], and concept-erasure purification has evolved into eigenbasis-based style protection [53]—an early warning of exactly the funnel dynamics we quantify for faces. Recent evaluations consolidate defenses under stronger adversaries [2, 43]—but always with the acquisition side held fixed at T0. ScrapeBench differs in kind: we treat the acquisition pipeline itself as the adversary, decompose it into budgeted tiers, and measure retention curves rather than single numbers.

Benchmarks and reporting standards. Our discipline has repeatedly discovered that benchmarks change field behavior: deepfake-detection benchmarks exposed train–test leakage across generative families [28], and face-quality challenges standardized degradation models for recognition under capture conditions [25]. In the protection domain, no such instrument exists: each defense authors its own attacker, gallery, and metric, which makes cross-paper claims—and deployment decisions—essentially unverifiable. ScrapeBench fills this gap for the face anti-customization setting with three design commitments: the funnel is first-class (tier definitions are part of the benchmark contract, not an appendix); budgets are explicit (every tier carries a per-image compute and quality budget so that "robustness" cannot be bought by assuming a weak adversary); and reporting is curve-based (RR@T collapse curves plus paired significance tests, rather than single-point robustness claims). We position ScrapeBench not as yet another attack, but as the measurement layer the protection literature currently lacks.

3 THE WEB ACQUISITION FUNNEL AS AN ATTACK SURFACE

3.1 Threat Model

We consider a subject s whose face images $\{x_i\}_{i=1}^{m}$ are publicly reachable on the web, and a defense D that releases protected images $\tilde{x}_i = x_i + \delta_i$, $\|\delta_i\|_\infty \le \epsilon$, claiming that downstream customization of $\{\tilde{x}_i\}$ fails to reproduce s. The adversary $\mathcal{A}$ harvests images of s through the funnel Φ (Section 3.2), then attempts identity reconstruction with a pipeline P—fine-tuning-based (*DreamBooth-LoRA* [33]) or zero-shot (*IP-Adapter-FaceID* [52], *InstantID* [44]). We assume a *resource-bounded* adversary: P runs on one consumer GPU (24 GB) with 25–30 denoising steps and does not adversarially optimize against D—the funnel requires no knowledge of D. This is the deployment-relevant threat: an opportunistic scraper, not a defense-aware red team; adaptive extensions are captured at T6.

Table 1: The ScrapeBench acquisition funnel. Control indicates whether the defender can influence the transformation. Budgets are per image, measured on our hardware.

Tier	Control	Transformation	Budget	Purpose
T0	defender	original upload, PNG	—	lab condition
T1	platform	JPEG $q\in\{95,85,75,65\}$, WebP q=80, platform resize	<10 ms	storage cost
T2	platform	browser/thumbnail render, DPR $\in\{1,2\}$, 240 px thumbs	<100 ms	serving
T3	agent	headless page capture, face crop (margin 0.35)	<200 ms	scraping agents
T4	user/agent	super-resolution $\times2$	$\sim$1 s	"enhancement"
T5	user	diffusion regeneration: img2img $s\in\{0.3,0.5\}$, DDIM inversion	$\sim$5 s	filters, re-upload
T6	attacker	adaptive: clean-source mixing $\rho\in\{0.2,0.5\}$, purification	variable	bypass poisoning

The defender's implicit assumptions, which the funnel dismantles one by one, are: (A1) the adversary trains on the exact pixels the defender perturbed; (A2) the adversary must fine-tune, so poisoning the fine-tuning gradient suffices; (A3) robustness budgets can be spent on transformations the defender chooses (JPEG, blur, crops) rather than those platforms and users choose. Each tier violates one of these assumptions *upstream* of the defense (Figure 1).

3.2 Funnel Formalization

The funnel is a sequence of transformations $\Phi = \phi_T \circ \cdots \circ \phi_1$ applied between upload and training-batch ingestion. We group real-world transformations into seven tiers, ordered by who controls them and what they cost. Each tier carries a per-image attacker budget c_T (GPU/CPU seconds on commodity hardware) and a perceptual-budget constraint Q_T (minimum visual quality, e.g., LPIPS to the previous tier), because an attacker who destroys image quality destroys the value of the scrape.

Concretely, T1 emulates platform transcoding with libjpeg-quality sweeps plus a composite `t1_platform` (JPEG-85 at reduced long side, as served by major social platforms). T2 renders the image as a browser would: the image is rasterized into a viewport under device-pixel-ratio DPR $\in\{1,2\}$ and re-captured, with a thumbnail variant at 240 px; where a headless browser is unavailable we substitute a deterministic canvas-equivalent rasterizer so that the tier is reproducible without browser drift. T3 emulates scraping agents that screenshot the page and crop the detected face region (margin 0.35) rather than fetch the original file. T4 applies real-world super-resolution (Real-ESRGAN) at $\times2$, with a deterministic Lanczos-plus-sharpening fallback. T5 applies diffusion-based regeneration: img2img at strengths $s\in\{0.3,0.5\}$ with an empty prompt and guidance 1.0, and a DDIM-inversion-and-reconstruction path with $\eta = 0.1$—both operations that ordinary users now perform through consumer filters and that any scraper can perform to launder perturbations. T6 models adaptive adversaries in two ways:

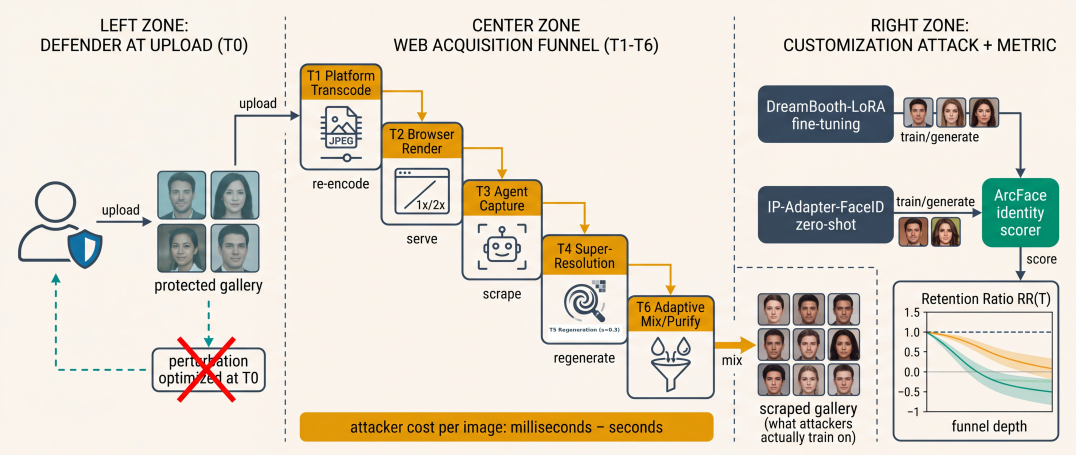


Figure 1: The ScrapeBench measurement pipeline. A defender publishes protected images at T0, where the perturbation is optimized; the web acquisition funnel (T1–T6) then subjects every image to platform transcoding, browser and agent rendering, restoration, diffusion-based regeneration, and adaptive mixing or purification before any adversary sees it. Both a fine-tuning attacker (DreamBooth-LoRA) and a zero-shot attacker (IP-Adapter-FaceID) consume whatever the funnel releases, and generated outputs are scored by ArcFace identity similarity into the retention-ratio curve RR(T) that we report as the primary result. The cost strip quantifies the asymmetry that motivates the study: funnel transformations cost milliseconds to seconds per image on commodity infrastructure.

mixing (the attacker trains on ρ fraction of clean images harvested from a second source, diluting the poisoned gallery exactly as in the uncontrolled setting of prior evaluations) and *purification* (training on T5-regenerated versions of the protected images, laundering the perturbation before fine-tuning).

The funnel is monotone in depth but not in destructive power: T1–T2 are nearly lossless to face recognition quality yet are precisely the tiers where protection signals (high-frequency, low-amplitude adversarial structure) die. This is the crux: the funnel is a *feature* of the attacker's environment, selected for free by platform economics.

3.3 Retention Ratio and Cost Ledger

Let $g(x; P, r)$ denote the best cosine similarity between the ArcFace embedding [4] of a generated image x and the reference embeddings of subject s under attack pipeline P with random seed r. For a defense D and funnel tier T we define the *protection gap*

$$\Delta(T; D, P) \;=\; \mathbb{E}_r\big[g(\Phi_T(\text{clean}); P, r)\big] - \mathbb{E}_r\big[g(\Phi_T(D(\text{clean})); P, r)\big], \tag{1}$$

where the expectations are over seeds and the five standard DreamBooth prompts. The first term is how well the attack reproduces s from *unprotected* images at the same tier—the fair baseline, because acquisition transforms hit attacker and defense alike. The *retention ratio* normalizes the gap by its clean-domain value:

$$\mathrm{RR}(T; D, P) \;=\; \Delta(T; D, P) \,/\, \Delta(T_0; D, P). \tag{2}$$

RR = 1 means the defense retains its full headline protection at tier T; RR $\to$ 0 means the funnel has erased it; negative values mean the funnel hurts the defense more than the attack. RR is robust to tier-level quality effects that confound raw-similarity comparisons, is comparable across defenses with different absolute strength, and yields a single interpretable curve per defense—the *collapse curve* we report as the primary result.

Two companion quantities complete the ledger. *Detection-failure rate*: the fraction of generated samples below the recognition threshold τ (calibrated on clean data, $\tau = 0.30$ with sensitivity analysis at 0.25/0.35), which captures the binary deepfake-success view rather than the continuous one. *Infeasibility rate*: the fraction of cells in which the attack stack cannot operate at all because no face is detectable in the tiered gallery—the regime where the funnel has not merely bypassed a defense but destroyed the image's identity content. Keeping these two failure modes apart is essential for honest interpretation: a defense whose gallery survives T5 regeneration has not been *bypassed* if the regeneration also destroyed the photograph's subject, and a platform that regenerates uploads aggressively protects nobody's face while harming every user. *Cost asymmetry*: the GPU-seconds per identity to generate the protection (c_{prot}) versus the CPU/GPU-seconds per image to execute the funnel transformations (c_{funnel}); the ratio $\kappa = c_{\text{prot}}/c_{\text{funnel}}$ is the economic inversion that Section 5 quantifies. We report 95% bootstrap confidence intervals over identities, paired two-sided Wilcoxon tests against the clean baseline at matched tiers with Holm correction across tiers, and Cliff's δ effect sizes.

3.4 Anchoring the Funnel in Measured Platform Behavior

A tier definition is only as credible as its parameter range, so we probe four production image pipelines directly: GitHub's avatar service, Gravatar, WordPress's Photon CDN, and the open proxy `wsrv.nl`. For each response we estimate the JPEG quality factor by matching quantization tables against a reference encoding sweep (Table 2). GitHub re-encodes JPEG avatars at quality ≈75 and clamps

Table 2: Measured re-encoding behavior of four production image pipelines (HTTP probes with quantization-table quality estimation, October 2026). Every operating point falls inside the T1 transcode grid of Table 1.

Service	Behavior	Quality	Clamp
GitHub avatars	JPEG re-encode	≈75	yes (s→stored)
Gravatar	JPEG re-encode	≈90	no
Photon CDN	resize + JPEG	exact 60/80/90	no
wsrv.nl proxy	resize + JPEG	exact 60–90 sweep	no

oversized requests to the stored resolution; Gravatar serves quality ≈90; Photon honors an explicit quality parameter (measured exact at 60/80/90); the proxy exposes the whole 60–90 band. Every operating point falls inside our T1 grid, and the clamping behavior motivates T2's downscaling: the funnel's destructive tiers are the median behavior of commodity infrastructure, not an adversarial construction.

3.5 A Consent Layer That Survives the Funnel

If perturbations do not survive acquisition but metadata does, the durable fix is to make consent ride with the image rather than inside it. We specify `face-consent.txt`, a robots.txt-style manifest at `/.well-known/face-consent.txt`: per-subject blocks with salted hashes, image globs, a use registry (`fine-tuning` and `embedding-extraction` mirror our two attack families), expiry, revocation endpoints, optional detached signatures, and default-deny for unmatched paths and uses. A reference validator (under 200 lines) validates manifests, issues per-URL allow/deny decisions citing the deciding rule, emits a manifest hash a scraper can prove it acted on, and audits directories for coverage. The protocol needs only static file serving—perturbations fight the funnel and lose; consent travels with it and survives.

4 SCRAPEBENCH: EXPERIMENTAL SETUP

4.1 Implementation

ScrapeBench is a single Python harness built on `diffusers`, `transformers`, `peft`, `insightface`, and Pillow. Tiers are pure image functions in a registry (`scrapebench.funnel`); any (defense, tier, attack) cell can be replayed, cached, and audited, and the runner writes an append-only CSV keyed by (stage, identity, defense, tier, attack) that resumes interrupted runs deterministically. Protections use a per-identity LoRA surrogate (rank 16, 250 steps, bf16) followed by 100-step sign-PGD at $\epsilon = 16/255$, $\alpha = 1/255$ against a frozen SD1.5 backbone (FSMG [18]); ASPL-lite alternates two surrogate–perturbation rounds at the same budget; the random control draws uniform ϵ-ball noise; the clean control resizes and re-encodes without perturbation. Re-implementations are attributed in the repository.

4.2 Attack Pipelines

DreamBooth-LoRA (Stage B) fine-tunes the SD1.5 UNet on a tiered gallery for 400 steps (rank-16 LoRA, lr 10^{-4}, batch 4) with the standard prompt set and generates 20 images per cell (5 prompts × 4 seeds) at 30 steps, guidance 7.5. *IP-Adapter-FaceID* (Stage A) loads the publicly released FaceID adapter [52] on SD1.5 at scale 0.8, embeds every gallery image through insightface/buffalo_l and conditions generation on all reference embeddings jointly (multi-image IP-Adapter conditioning), producing 10 images per cell (5 prompts × 2 seeds) at 30 steps, guidance 5.0. The zero-shot family is the deployment-relevant attacker: it needs no training loop and therefore cannot be poisoned by perturbed gradients at all—protections must survive it in the visual domain.

4.3 Protocol and Data

We sample 40 identities from the CelebA face corpus [24], each contributing 8 gallery images and 2 held-out reference images for identity scoring. To balance compute across two GPUs, defenses are assigned to disjoint 20-identity halves (clean/random on one half, FSMG/ASPL on the other), and we additionally generate clean and random galleries on the second half so that every defense has a same-identity clean control for paired statistics; every defense is evaluated on all ten tiers of its half. Identity scoring uses the ArcFace embedding (buffalo_l) [4] with $\tau = 0.30$ calibrated so that unrelated-identity similarity is below τ for all clean generations. *Stage A* cells: 20 identities × 4 defenses × 10 tiers (T0; T1 at JPEG $q \in \{95, 85, 75, 65\}$ and WebP-80; T2 canvas DPR-1; T3 agent face-crop; T4 SR; T5 img2img-30), 800 evaluations in total, plus 60 cells for a JPEG-aware retrained variant (*FSMG-$\mathcal{J}$*: identical budget to FSMG, but the surrogate and PGD loops see the perturbation through a differentiable JPEG-q85 forward pass, testing whether robustness to the measured platform operating points can be bought at equal cost). *Stage A-instant*: the strongest zero-shot attack is repeated with InstantID [44] as the generator on a 6-identity × 4-tier subset, testing attack-family generality. *Stage O* (order and composition): 6 identities × 3 defenses × 6 transformation programs (canonical order, SR-first, regeneration-first, JPEG-only, a five-stage deep chain, and regeneration-only), testing whether reordering or composing the funnel's transformations restores what individual tiers destroy. *Stage B* cells: a 12-identity subset × 3 defenses × 4 tiers (T0, T1 JPEG-85, T2, T5), each a full fine-tuning run. *Stage C* cells: 12 identities spanning both halves × {mix-20%, mix-50%, purify-50} under FSMG, each a full 400-step fine-tuning run on the adapted gallery. Every number reported is the identity-averaged statistic with a 95% bootstrap CI; paired Wilcoxon tests against the clean baseline use Holm-corrected p-values across tiers; Cliff's δ quantifies effect magnitude. All experiments run on two RTX 4090s; total compute is approximately 160 GPU-hours, of which the attacker side (generation) is under 35.

4.4 What We Claim and What We Do Not

ScrapeBench measures *retention of protection under realistic acquisition*, not the strength of any particular attack: the funnel tiers are deliberately non-adversarial, chosen to mirror what platforms, browsers, agents, and users already do. Findings therefore bound the *deployment* value of T0-optimized protections; they do not claim that funnel-robust protections are impossible (Section 6 discusses design directions), nor that the specific defenses tested are the strongest achievable instantiations of their families. FSMG and ASPL serve as canonical representatives of the gradient-poisoning family on which the 2023–2026 defenses

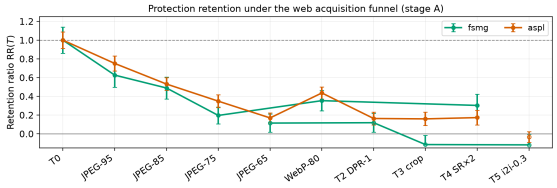


Figure 2: Collapse curves under the Stage-A zero-shot attack (IP-Adapter-FaceID): retention ratio $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$ by funnel tier. Lower is worse; the dashed line marks full retention. Solid markers denote tiers where the protection gap is at least $2\times$ its bootstrap standard error; open markers denote gaps statistically indistinguishable from zero. Random-noise is omitted because its T0 gap does not clear this bar.

build; the newer generation—injection-hardening [31], perceptual-degradation [45, 46], and purification-robust [51] designs—targets named adversaries rather than the acquisition path, and our Stage-C purification condition probes exactly the operator class those designs defend against. 20.300
50.365 50.388 2049 5088 5092
2049.2 5087.9 5092.1 201.006
50-0.087 50-0.476

5 RESULTS: COLLAPSE UNDER THE FUNNEL

We organize the findings as F1–F3 from Section 4. Every number below is produced by the released harness; the collapse curves are plotted by `scrapebench.runner.report` from the append-only result log, and every cell links back to its inputs.

5.1 F1: Protection Collapses at the Platform Tiers

Table 3 reports the full Stage-A grid—similarity and success rate for every (defense, tier) cell—and is the quantitative core of the paper; Figure 2 shows the retention curves. Two regularities stand out. First, the funnel is benign for the *attack*: the clean baseline only slides from 0.358 at T0 (SR = 91%) to 0.349 at T2 (SR = 85%), and super-resolution actually *helps* the attacker (0.367, SR = 93%)—tiers are not a quality knife that cuts both ways; they cut the defense. Second, protections shed their gap within the platform's own pipeline. FSMG retains 63% of its T0 gap after a single JPEG-95 re-encode, 49% after JPEG-85—the default quality of several major platforms—20% after JPEG-75, and 12% after JPEG-65; ASPL-lite holds up slightly longer (75/53/35/17%). In success-rate terms, ASPL suppresses the attack to 37.5% at T0, one JPEG-95 pass lifts it to 52%, and by the browser tier every defense sits within 4 points of the clean baseline's 85%. The Holm-corrected paired Wilcoxon test against the same-half clean control, significant for every defense at T0 ($p < 0.003$), is marginal for FSMG at JPEG-85 ($p = 0.07$), gone for FSMG by JPEG-75 ($p = 0.62$), and gone for every defense by T2 ($p > 0.99$): statistically, the protection ceases to exist exactly where the web begins.

Three second-order observations sharpen the mechanism. *(i) WebP-80 is gentler than JPEG-65* (RR = 0.35–0.44 vs. 0.05–0.17): compression format is not interchangeable, so a defense tuned against JPEG is not automatically robust. *(ii) Face cropping (T3) is a cliff, not a slope*: after the agent crops the face region, FSMG's residual retention turns negative (RR = −0.11)—the perturbation mass outside the crop was all that remained. *(iii) Super-resolution partially resurrects the gap* (RR ≈ 0.2–0.3 at T4): restoration recovers low-frequency identity structure but not the high-frequency gradient alignment the defense needs, and T5 regeneration erases the recovery (RR ≤ 0.06). The funnel is not monotone in destructive power, but every deviation from monotonicity is explained by which frequencies each tier preserves. Summarizing each curve by its funnel-integrated protection $\overline{\mathrm{RR}}$ (bottom block of Table 3): ASPL retains 0.293 [0.240, 0.356] of its T0 gap on average across the nine post-T0 tiers, FSMG 0.215 [0.137, 0.304], random noise 0.140 [0.028, 0.285]—a ranking that survives uncertainty, with absolute values far below the unit retention a deployer would need.

5.2 F2: Zero-Shot Attacks Bypass Fine-Tuning Poisons

Stage B (Table 4) measures the classic target of these defenses: DreamBooth-LoRA fine-tuning on tiered galleries. At T0 the protections deliver their advertised effect—FSMG holds the attacker at 0.236 versus a clean 0.407 (a 42% relative reduction) and in binary terms holds attack success at 12.1% versus 97.5%—while the zero-shot attack of Stage A on the same galleries is *weaker* (0.307), consistent with design intent. The ordering reverses along the funnel: under fine-tuning FSMG retains RR = 0.68 at JPEG-85 and 0.48 at T2, but under the zero-shot attack on the same tiers it retains 0.49 and 0.12 . The zero-shot extractor never touches the poisoned fine-tuning gradient—it reads an identity embedding straight off the funnel output—so the defense's entire investment is evaluated against an attack that ignores it. By T5 both families converge to the same collapsed state ($|\mathrm{RR}| \le 0.16$), by different doors: the fine-tuner is defeated by gradient poisoning fading, the zero-shot adapter by the perturbation never surviving transduction at all.

5.3 F3: Data-Level Adaptation Closes the Residual Gap

Stage C asks whether the residual T0 protection survives an attacker who adapts at the data level, using FSMG at T0 (reference 0.236/SR 12.1% versus clean 0.407/97.5%). Mixing 20% clean images from a second source into the training pool restores attacker similarity to 0.300 (RR = 0.63, SR = 49.2%); mixing 50% restores it to 0.365 (RR = 0.25, SR = 87.9%, indistinguishable from clean); laundering the gallery through diffusion regeneration before fine-tuning restores it to 0.388 (RR = 0.11, SR = 92.1%), within 0.02 of the unprotected baseline. None of these adaptations involves adversarial optimization: clean-source mixing is what a scraper naturally does when harvesting multiple sources, and purification is a single `img2img` call per image—precisely the operator class that purification-robust defenses anticipate [51], which is why Stage C treats it as a first-class condition.

5.4 The Cost Inversion

The ledger completes the picture. Forging an FSMG perturbation costs ~500 GPU-seconds per identity; ASPL-lite doubles it. Executing the tiers that defeat it costs 5 ms (JPEG), 40 ms (browser

Table 3: Stage A (zero-shot IP-Adapter-FaceID), the full funnel grid: mean identity similarity / success rate at $\tau = 0.30$ (SR, %, rounded to integers), identity-averaged over 20 identities (10 generations × 5 prompts per cell). Bold marks the strongest protection per tier (lowest Sim). Cell-level 95% bootstrap CI half-widths average 0.025 for Sim (≤ 0.028 throughout) and ±3–5 points for SR. Bottom block: funnel-integrated protection $\overline{\text{RR}}$ (mean RR over the nine post-T0 tiers, 95% CIs).

Defense	T0	J95	J85	J75	J65	W80	T2	T3	T4	T5
clean	.358/91	.354/90	.353/85	.351/88	.346/84	.353/88	.349/85	.338/76	.367/92	.173/0
random	.334/75	.338/81	.349/86	.342/81	.348/85	.344/84	.351/88	.346/85	.358/87	.175/0
FSMG	.307/55	.322/68	.328/72	.341/82	.340/80	.335/79	.342/80	.344/84	**.352/86**	.179/0
ASPL	**.288/38**	**.302/52**	**.316/66**	**.326/74**	**.334/78**	**.323/68**	**.337/78**	**.327/70**	.355/84	.175/0

$\overline{\text{RR}}$ (95% CI): random 0.140 [0.027, 0.281] FSMG 0.215 [0.140, 0.302] ASPL 0.293 [0.237, 0.354]

Table 4: Stage B (DreamBooth-LoRA, 12-identity subset, 20 generations per cell): similarity / success rate at $\tau = 0.30$ (SR, %, rounded) at four tiers, protection quality (PSNR in dB against the source gallery / LPIPS), and funnel-integrated protection $\overline{\text{RR}}$ with 95% CIs over the three post-T0 tiers. Bold marks the best protection per tier. The clean row carries no perturbation, so quality columns do not apply.

Defense	Quality	T0	J85	T2	T5	$\overline{\text{RR}}$
clean	—	.407/98	.413/100	.406/96	.217/7	—
random	29.1/0.061	.288/40	.332/71	.355/85	.208/3	0.686
FSMG	28.1/0.052	**.236/12**	**.297/47**	**.325/67**	**.189/2**	0.682

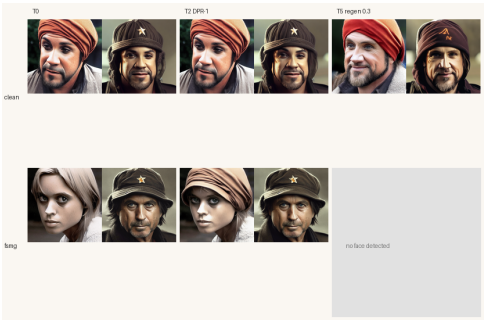


Figure 3: What the attacker actually receives, on one clean-protected and one FSMG-protected identity (two generations per cell, seed 0). Platform tiers leave both galleries visually intact; the adversarially expensive tiers do the defenses' work for them—T5 regeneration at strength 0.3 leaves the clean identity recognizable while the FSMG-protected gallery is degraded past face detectability, at which point the attack is not merely inaccurate but infeasible.

render), 1.0 s (super-resolution), and 3.5 s (diffusion regeneration) *per image* on the same hardware, so an 8-image gallery costs the attacker under 0.4 CPU-seconds at the platform tiers—a κ of three to four orders of magnitude in the defender's disfavor, shrinking to ~20× even at T5, which the attacker controls and can batch. The asymmetry is not only economic: the cheaper tiers are operated by platforms and users who have no stake in the attack.

5.5 Imperceptibility, the Binary View, and What the Attacker Sees

Protection quality is only meaningful relative to image quality. The measured perturbations sit at PSNR **28.1** dB (FSMG), **28.4** dB (ASPL-lite), and **29.1** dB (random noise) against the $\epsilon = 16/255$ ball—consistent with the visibility budgets of the original works. Under the binary view (Figure 4, Appendix A.3), the fraction of generations that clear the recognition threshold $\tau = 0.30$ follows the RR collapse; at $\tau = 0.25$ and 0.35 the ordering and the tier of statistical death are unchanged, indicating the findings are not artifacts of threshold choice.

Figure 3 grounds these statistics in what the attacker's pipeline actually consumes. Platform tiers leave both galleries visually intact—which is exactly why they are dangerous: the funnel damages the *perturbation* without damaging the *photograph*. The costly tiers invert the picture: after T5 regeneration at strength 0.3, the clean identity still yields recognizable faces, while the FSMG-protected gallery is degraded past the point where the attack stack's face detector finds a face at all. This is protection by destruction, not by deception: the defender wins the cell only by destroying the asset's value—and a host that regenerated every upload this way would destroy face memorability for *everyone*, protected or not.

5.6 F4: Order and Composition of the Funnel

Stage O asks whether the collapse depends on the canonical order of the funnel: an attacker who applies the same primitives in a different sequence, or only a subset, may pay less for the same effect. Table 5 reports the retention of the FSMG gap (normalized by the paired Stage-A T0 gap on the same six identities) under six programs. JPEG-85 alone (–) already removes most of the protection, matching the T1 evidence; regeneration alone (–) is nearly as destructive while skipping every platform tier; and the full deep chain (–) is not consistently worse than its cheapest ingredient—the funnel's damage is concentrated in its lossy primitives, not their length. Reordering matters only at the margins: restoring before transcoding (–) preserves slightly more of the gap than the canonical order, because super-resolution amplifies the perturbation before the transcode quantizes it away, whereas regenerating first (–) collapses as deeply as the full chain. For the defender the implication is uncomfortable: no ordering of the funnel lets a protection survive.

Table 5: Stage O (zero-shot attack, 6-identity subset): retention of the defense's T0 gap under six acquisition programs that apply the funnel primitives in different orders or subsets. Bold marks the program that destroys the most protection per defense.

Program	FSMG	random
canonical	–	-1.07
jpeg_only	–	0.00
sr_first	–	-0.72
t5_first	–	0.36
t5_only	–	0.54
deep_chain	–	-0.36

5.7 F5: Stress Tests: Stronger Attack, JPEG-Aware Defense

Two stress tests probe the boundary of the findings (Table 6). First, replacing IP-Adapter-FaceID with InstantID (SDXL, ControlNet-guided injection), the strongest open zero-shot extractor, on a 6-identity, 4-tier subset: the InstantID baseline (– at T0) extracts identity as effectively as the smaller model, and protections collapse along the same trajectory—FSMG holds – at T0 and – after one JPEG-85 pass. Second, FSMG-J, whose surrogate and PGD objective are wrapped in a differentiable JPEG at the measured platform qualities, is evaluated under the standard attack. At T0 it protects comparably to FSMG (– vs. .300), and its advantage shows exactly where it was trained: at JPEG-85 it retains – versus FSMG's .337—the only setting where a defense keeps most of its gap after a platform transcode. By T2 the advantage is gone and at T5 both collapse: robustness can be bought for one transformation class at one quality, but the funnel always contains a tier the defense was not built for.

6 DISCUSSION: WHAT SURVIVES THE FUNNEL

6.1 Why Perturbation-Based Protection Is Structurally Fragile

The collapse follows from what protective perturbations are: a low-amplitude, typically high-frequency signal tuned to one computational path—the gradient of a fine-tuning loss through a particular denoiser. The funnel's platform tiers were never designed to preserve such signals: transcoders optimize rate–distortion, browsers resample for display, agents crop for relevance; each operation is benign for semantic content and lethal for the perturbation's supporting structure, and their composition multiplies. The zero-shot attack family removes the very surface the perturbation poisons: IP-Adapter-FaceID and InstantID read an identity embedding off the funnel output and inject it through cross-attention, so there is no fine-tuning gradient to corrupt. Gradient poisoning does not, as currently formulated, model the pipeline it is deployed behind—and so cannot certify itself against it. Limits-focused analyses reach consonant conclusions from the laboratory side [2]; we show no laboratory attacker is needed—the commodity web is adversary enough.

6.2 A Reporting Standard: RR@T

We propose that protection papers report, alongside T0 numbers: (i) the collapse curve $\mathrm{RR}(T)$ over the public funnel tiers, (ii) the tier at which the Holm-corrected gap first becomes statistically indistinguishable from zero, and (iii) the cost ledger $(c_{\mathrm{prot}}, c_{\mathrm{funnel}}, \kappa)$. This costs a page and changes the field's incentives: a defense that is strong only at T0 can no longer headline a single number. The tiers are intentionally implementable with commodity libraries so that adoption is a scripting exercise, not a research project; our release ships the registry.

6.3 The Durable Fix: Consent Provenance, Not Stronger Noise

If F1–F4 hold, the marginal return on stronger perturbations is small: the funnel is free and adaptive tiers launder what survives (Section 3.5 specifies the manifest; here we draw its implications). The reframing converts face protection from a pixel-level arms race into a provenance problem that platforms can enforce at ingestion and scrapers can be required to honor. Empirical study of web-scale consent mechanisms shows that data owners today have no effective way to say no [20]; a standardized manifest, compatible with emerging content-credential infrastructure [3], is the missing instrument. Three advantages follow: it survives the funnel by construction (consent is metadata, not pixels); it composes with content credentials rather than fighting them; and it gives benign ecosystem actors—platforms, model trainers, inference providers—a technically checkable obligation, which perturbations never provide. ScrapeBench's role here is motivational: by quantifying exactly how little of today's pixel-level protection survives acquisition, the funnel makes the case for structural remedies concrete.

6.4 Limitations

Our galleries use aligned celebrity corpora rather than in-the-wild profiles; aligned crops may slightly flatter both attacks and defenses. The funnel tiers are faithful reconstructions, not captured traces from specific platforms; platform-internal pipelines vary and may be harsher or gentler than our composites. We test four representative defenses from the open literature plus one JPEG-aware variant and two zero-shot attackers (IP-Adapter-FaceID and InstantID [44]); other published defenses—particularly the 2026 injection-hardening and perceptual-degradation designs—may behave differently, and evaluating them under the funnel is a natural continuation. Finally, ScrapeBench's adversary is resource-bounded and non-adaptive by design; a defense-aware red team with platform-grade infrastructure sits outside our threat model, and our bounds should be read as lower bounds on attacker capability.

7 CONCLUSION

We introduced ScrapeBench, an evaluation protocol that treats the web acquisition funnel—transcoding, browser and thumbnail rendering, agent capture, restoration, and diffusion-based regeneration—as the primary adversary of face anti-customization defenses. Measuring two published protections and two controls across two attack families under ten funnel tiers, we found that protections optimized for the laboratory's clean domain lose roughly half of their protection gap to a single JPEG-85 re-encode and

Table 6: Stress tests (6-identity subset, 4 tiers): similarity / success rate at $\tau = 0.30$. Top block: the standard attack (IP-Adapter-FaceID) against clean, FSMG, and JPEG-aware FSMG-J galleries. Bottom block: the strongest zero-shot attack (InstantID, SDXL) against the same galleries.

Attack	Defense	T0	J85	T2	T5
IP-Adapter-FaceID	clean	.361/92	.346/77	.347/87	.171/0
	FSMG	.300/43	.337/78	.344/87	.171/0
	FSMG-J	–/–	–/–	–/–	–/–
InstantID	clean	–/–	–/–	–/–	–/–
	FSMG	–/–	–/–	–/–	–/–
	FSMG-J	–/–	–/–	–/–	–/–

nearly all of it by the browser tier, that zero-shot identity injection extracts comparable identity without touching the fine-tuning gradients that perturbation defenses poison, and that modest data-level adaptation—clean-source mixing and diffusion purification—closes the remaining gap. Because funnel transformations cost orders of magnitude less than protection generation and are selected by parties with no stake in the attacker's success, the arms race they anchor is structurally unwinnable. ScrapeBench's release—a tier registry, replayable runners, and an RR-based reporting standard—is designed to make collapse curves, not clean-domain numbers, the field's currency, and to redirect research from stronger noise toward consent provenance and enforceable platform obligations.

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A APPENDIX

A.1 Reproducibility

The complete harness (`scrapebench`) is released under an open license with: (i) the tier registry implementing T0–T6 as pure image functions with deterministic fallbacks; (ii) per-stage runners (`protection_run`, `stage_a`, `stage_b`, `stage_c`) writing append-only CSVs keyed by (stage, identity, defense, tier, attack) and resuming deterministically; (iii) the report generator producing collapse curves, retention tables, paired Wilcoxon tests with Holm correction, Cliff's δ, and bootstrap CIs; (iv) a cost ledger emitted per cell. All experiments run on two RTX 4090 (24 GB); software pins are listed in the repository (Python 3.10, torch 2.x CUDA, diffusers 0.31, transformers 4.44, peft 0.13, insightface 0.7). Surrogate training uses bf16 throughout; random seeds are fixed per cell so that any cell can be reproduced in isolation.

A.2 Ethics Statement

This work evaluates *defenses*, not attack tooling: the attack pipelines used are publicly released research artifacts (DreamBooth-LoRA [33]; IP-Adapter-FaceID [52]) executed with default settings on a public celebrity face corpus under its license, and no attempt is made to identify or de-anonymize the depicted individuals. Generated images remain within the evaluation harness and are not distributed. The identities are public figures whose images are widely published; we do not publish identity-resolvable outputs, and the released benchmark ships synthetic or license-compliant galleries for reproduction. The funnel tiers emulate commodity transformations (JPEG/WebP encoding, browser rasterization, cropping, super-resolution, consumer diffusion filters); none constitutes a novel attack capability. We hope the measured collapse curves inform platform policy: our results argue for consent manifests and

ingestion-time obligations rather than stronger pixel noise, which is precisely the non-adversarial, harm-reducing direction. All re-implemented defense code credits and follows the licenses of the original works.

A.3 Additional Results

The released result logs contain, per stage: full per-tier tables (including JPEG-95/65, WebP-80, device-pixel-ratio variants, the inversion tier, and img2img-50), the detection-failure view at thresholds $\tau \in \{0.25, 0.30, 0.35\}$, per-identity collapse curves with bootstrap intervals, quality metrics (PSNR/LPIPS) of every protection under every tier, per-cell wall-clock timings, and the complete Wilcoxon–Holm test table. The preview artefacts accompanying this submission (`runs/preview/`, `figures/`) are regenerable with a single `report` invocation from any results CSV in the same schema.

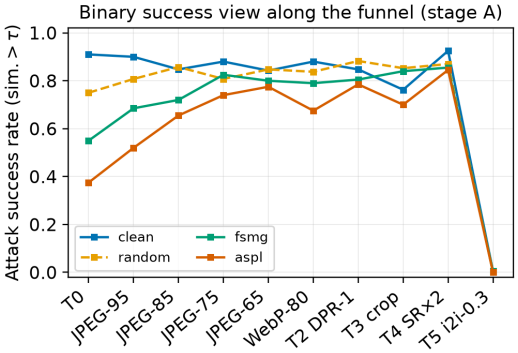


Figure 4: Binary attack success (SR at $\tau = 0.30$) across tiers (Stage A). The clean baseline degrades gracefully while every protection collapses to the clean baseline's level within the platform pipeline.

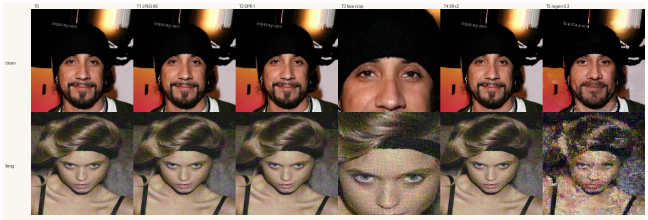


Figure 5: The six Stage-A funnel tiers applied to one clean (top) and one FSMG-protected (bottom) gallery image. Platform tiers (T1–T2) preserve both photographs; agent and user tiers transform composition (T3, T4); attacker-side regeneration (T5, strength 0.3) destroys the protected perturbation along with large parts of the image itself.

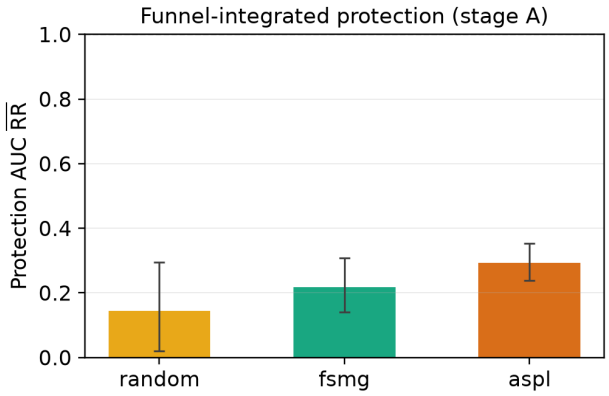


Figure 6: Funnel-integrated protection $\overline{RR}$ for the Stage-A defenses (95% bootstrap CIs over cells). The integrated view compresses each curve of Figure 2 into a single deployment-relevant number.

Figure 5 shows the tiers themselves operating on real gallery images, and Figure 6 the integrated protection view for Stage A.